

```
In [2]: # oads either a stack or map, including survey maps for analysis
# Browse to the file
# Summary of the stack shown in console
sgm_data = analyze_sgm_bsky_data()
```

Analyzing File: D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_stack\2026-04-24_223037_stack_data.h5

--- Scan Analysis Summary ---

Energies (182 points): [395. 395.5 396. 396.5 397. 397.5 398.5 398.6 398.7 398.8 398.9 399.

399.1 399.2 399.3 399.4 399.5 399.6 399.7 399.8 399.9 400. 400.1 400.2
 400.3 400.4 400.5 400.6 400.7 400.8 400.9 401. 401.1 401.2 401.3 401.4
 401.5 401.6 401.7 401.8 401.9 402. 402.1 402.2 402.3 402.4 402.5 402.6
 402.7 402.8 402.9 403. 403.1 403.2 403.3 403.4 403.5 403.6 403.7 403.8
 403.9 404. 404.1 404.2 404.3 404.4 404.5 404.6 404.7 404.8 404.9 405.
 405.1 405.2 405.3 405.4 405.5 405.6 405.7 405.8 405.9 406. 406.1 406.2
 406.3 406.4 406.5 406.6 406.7 406.8 406.9 407. 407.1 407.2 407.3 407.4
 407.5 407.6 407.7 407.8 407.9 408. 408.1 408.2 408.3 408.4 408.5 408.6
 408.7 408.8 408.9 409. 409.1 409.2 409.3 409.4 409.5 409.6 409.7 409.8
 409.9 410. 410.1 410.2 410.3 410.4 410.5 410.6 410.7 410.8 410.9 411.
 411.1 411.2 411.3 411.4 411.5 411.6 411.7 411.8 411.9 412. 412.1 412.2
 412.3 412.4 412.5 412.6 412.7 412.8 412.9 413. 413.1 413.2 413.3 413.4
 413.5 413.6 413.7 413.8 413.9 414. 414.1 414.2 414.3 414.4 414.5 414.6
 414.7 414.8 414.9 415. 415.5 416. 416.5 417. 417.5 418. 418.5 419.
 419.5 420.]

Number of Images: 182

Energy Regions: 395.00-397.50: 0.5 eV, 398.50-415.00: 0.1 eV, 415.50-420.00: 0.5 eV

X array (shape): (289,) (Nx: 17)

Y array (shape): (289,) (Ny: 17)

Grid Dimensions: 17 x 17

 Date: 2026-04-24

Scan Name: Au13_0013_n1s

Project: 40G14650

Beamline: N/A

Polarization: N/A

Grating: N/A

Harmonic: N/A

Strip: N/A

Command: N/A

Coordinates: X: 3.75 to 5.27, Y: -2.90 to -1.84

Exit Slit Gap: N/A

MCC Files Found: 182

SDD Files Found:

Detector sdd1: 182 files

Detector sdd2: 182 files

Detector sdd3: 182 files

Detector sdd4: 182 files

In [3]:

```
# Allows for the alignment of the stack to be checked
# Allows for removing some pixels ( as mm units) on the X and Y edges, which makes the stack dimensions smaller
# The new dimensions will be used in the plotting and PCA/CA
# This step can be skipped
interactive_roll_align(sgm_data, (30,50), use_color=False)
```

✓ 0.4s

Detector: Energy (eV):

Roll Shift:

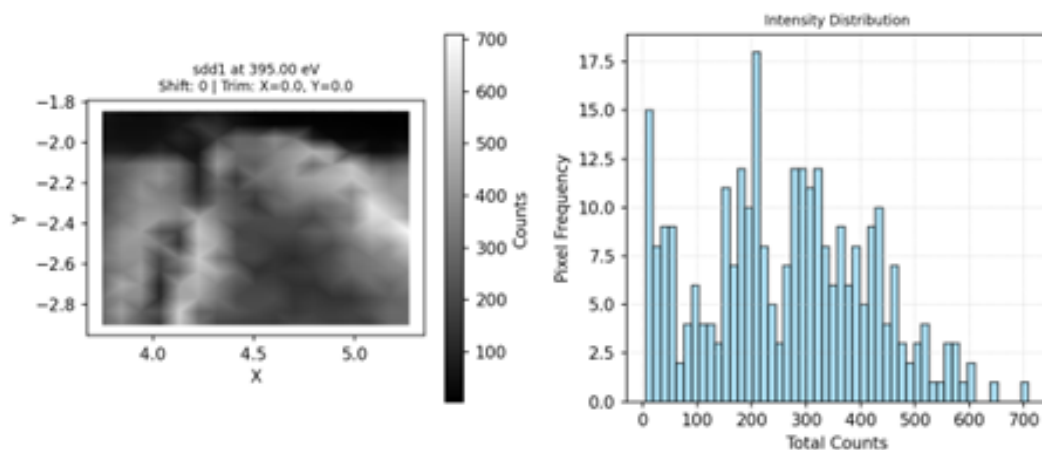
X-Trim (mm):

Y-Trim (mm):

☐ Use Color

Ref Energy:

align_fig_2575307757184



```
# Maps the fluorescence and sample TEY signal. Can select which image for the map.
# Plots the fluorescence and TEY spectra from a stack- saves Raw and Normalized
XANES spectra for each SDD, I0 and sample TEY as a csv file.
# Saves XRF spectra for each SDD as a csv file. Saves images (i.e., maps) for each
SDD.
```

```
# Can change the ROI from the XRF spectrum to select the edge of interest.
# Can change the area selected on the image by dragging a rectangle or making a
polygon over the area of interest. Can toggle between the two. Note that the area
selected will be where the spectra are derived from and saves in the csv file.
# To see the changes in the spectra need to click on the "Refresh Plots" button.
# Can use the internal I0 from the Au mesh (mcc1) or load an external I0 from a csv
file for normalization- allows for which column to select.
```

```
# MCC1 is the I0 from Au mesh before the KB mirrors, MCC2 is from a photodiode behind the
sample (only collected when no sample in the pathway), which could be used as an I0, MCC3
auxillary, and MCC4 is the TEY from the sample.
```

```
# Can either show lines with or without individual data points.
```

```
# Can zoom in on spectra using tools o
# Mouse over an image or plot and get the X and Y coordinates.
# Automatically adjusts the contrast for each image.
# Plot as a scatter map (usual way) or as a heatmap
# The default ROI on the XRF spectra is 0-255. Drag Left and right on the XRF spectra to
select the ROI for the element of interest. Each bin is about 10 eV.
# Calibrate the energy by adding a energy shift value - energy_shift = 0 as default.
```

```
# Can display spectra with the data points or not- show_markers = False spectra plotted
without data points shown.
```

```
# Can change the display of the images, selecting grey scale or color
```

```
plot_sgm_bsky_data(sgm_data, as_scatter_plot=True, fixed_roi=False, mcc_to_map=[4],
show_markers=False, mcc_channels=[1,4], energy_shift=0)
#plot_sgm_bsky_data(sgm_data, as_scatter_plot=True, use_color=False, fixed_roi=False,
channel_roi=[30,50]
```

```
[Auto Select] Defaulting to middle energy: 407.00 eV
--- SDD Stack Dashboard: Au13_0013_n1s ---
[Metadata] Full Headers: Disabled
Location: D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_st
ack
```

```
[Alignment] Roll Shift: 0 | Trim: X=0.000 mm, Y=0.000 mm
Pass 1: Pre-loading energy dependence...
```

[ROI ENGINE v2.6 Ready] Dashboard Active.

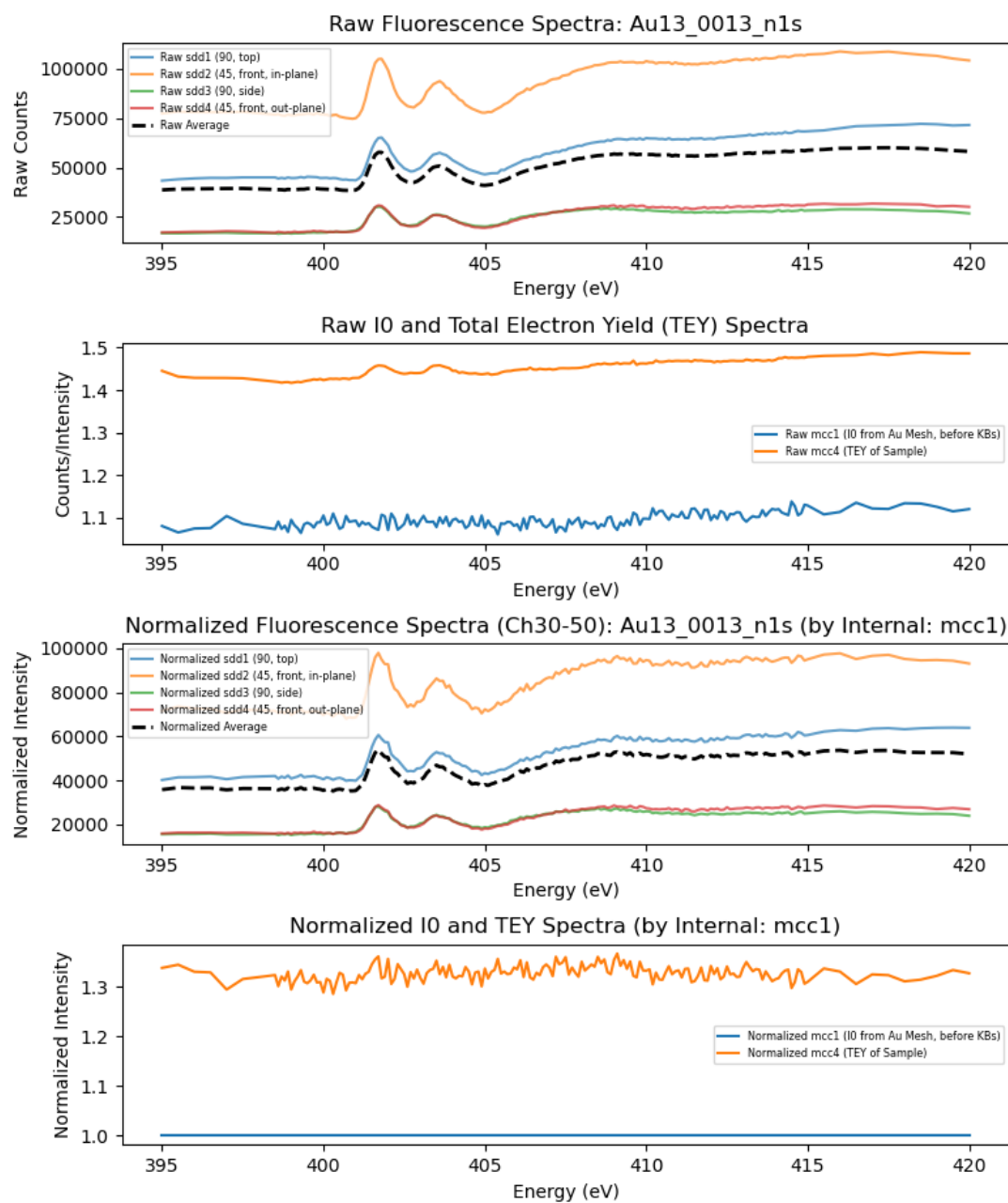
REFRESH PLOTS

dash_summary_2576196929920



Image Energy Selector

91

☐ Add Sample Specific Information to File Header

Save XANES Spectra

Save XRD Spectra

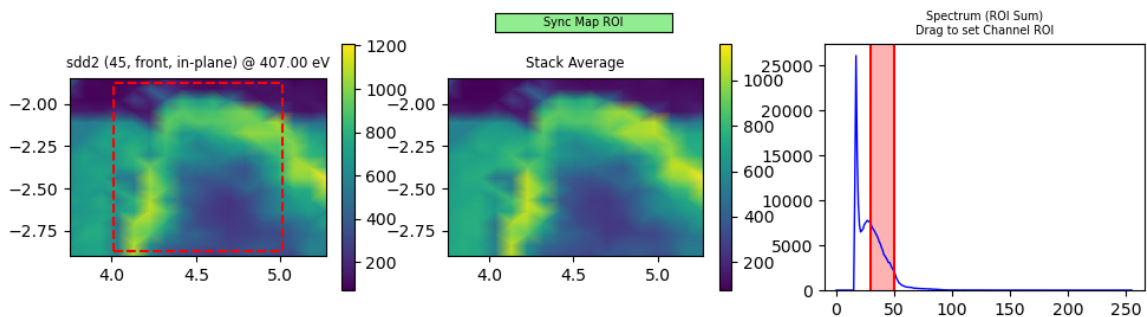
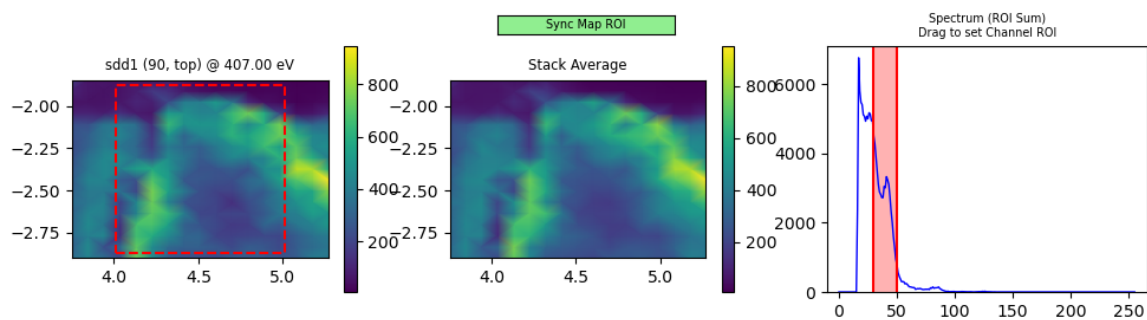
Switch Polygon ROI Selection

Switch to Gray

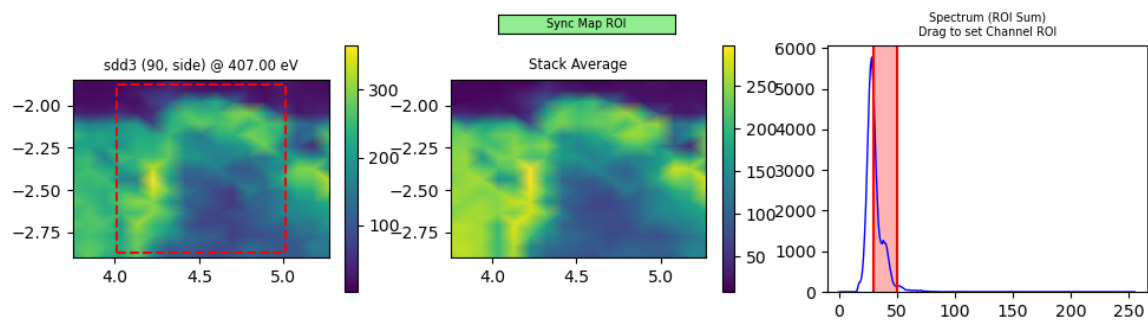
Save Images

Out[]: <plot_sgm_bsky_data.Synchronizer at 0x257d14d8980>

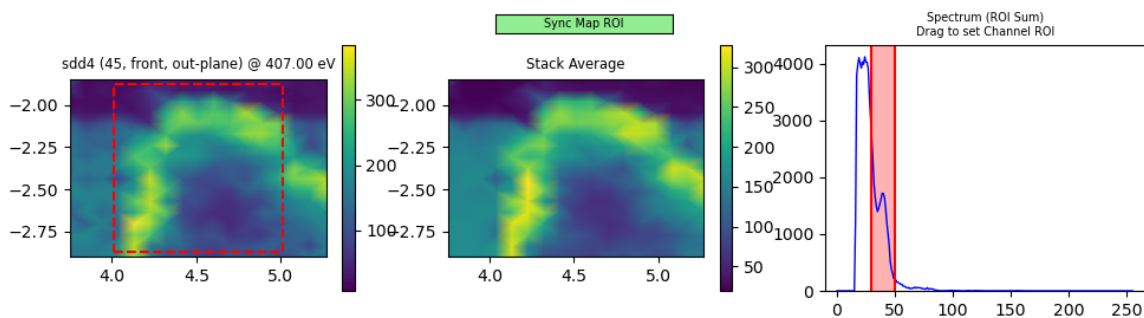
dash_sdd1_2576196929920



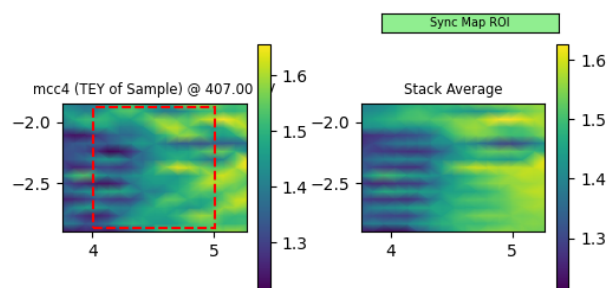
dash_sdd3_2576196929920



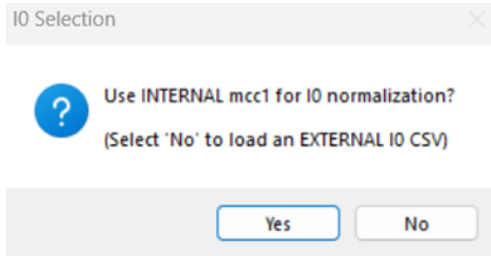
dash_sdd4_2576196929920



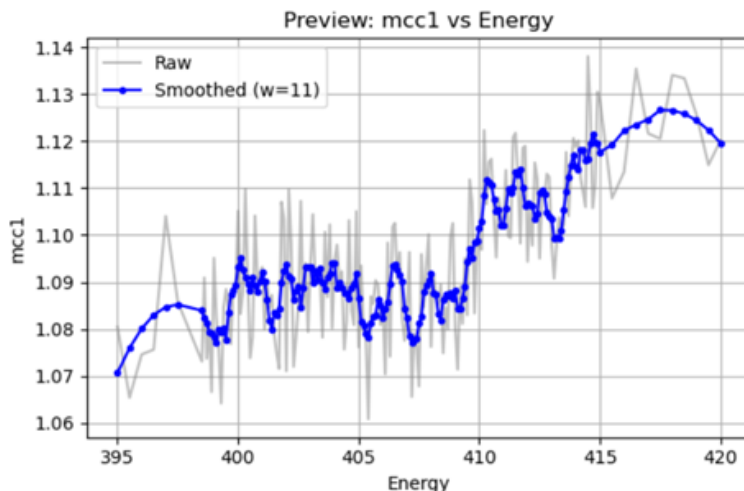
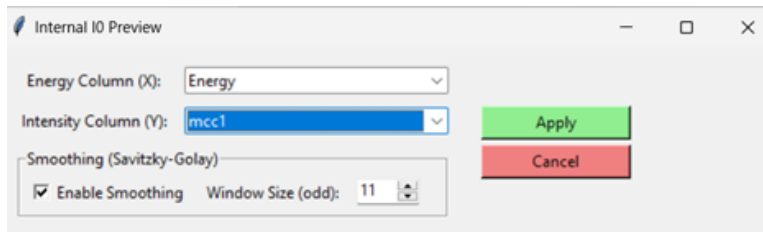
dash_mcc4_2576196929920



When first loading the stack you will be asked whether to use the internal I0 from the current stack or load an external I0 determined from another stack for the normalization. Selecting the external I0 you will select a column from a previously saved csv file.



After selecting either an internal or external I0 a gui will open that allows you to smooth the I0 by checking the enable smoothing box. Then determine the amount of smoothing desired. Once satisfied, apply.



The header of saved spectra will routinely be saved with the following information. Check the box to save specific sample information as shown in the bottom right hand corner on this page.

```
# Scan Name: Au13_0013_n1s
# Project: 40G14650
# Date: 2026-04-24
# Number of Images: 182
# Energy Regions: 395.00-397.50: 0.5 eV,
398.50-415.00: 0.1 eV, 415.50-420.00: 0.5 eV
# Grid: 17 x 17 (Total: 289)
# Grating: N/A
# Harmonic: N/A
# Strip: N/A
# Polarization: N/A
# Exit Slit Gap: N/A
# ROI Selection: Rect
# Channels: Ch30-50
# Normalization: Internal: mcc1
#
# Column 1: Calibrated Energy (eV)
# Column 2: Original Energy (eV)
# Column 3: RAW_sdd1 (90, top)
# Column 4: RAW_sdd2 (45, front, in-plane)
# Column 5: RAW_sdd3 (90, side)
# Column 6: RAW_sdd4 (45, front, out-plane)
# Column 7: RAW_Average_SDD
# Column 8: NORM_sdd1 (90, top) (by Internal: mcc1)
# Column 9: NORM_sdd2 (45, front, in-plane) (by
Internal: mcc1)
# Column 10: NORM_sdd3 (90, side) (by Internal: mcc1)
# Column 11: NORM_sdd4 (45, front, out-plane) (by
Internal: mcc1)
# Column 12: NORM_Average_SDD
# Column 13: RAW_mcc1 (I0 from Au Mesh, before KBs)
# Column 14: RAW_mcc4 (TEY of Sample)
# Column 15: NORM_mcc1 (I0 from Au Mesh, before KBs)
(by Internal: mcc1)
# Column 16: NORM_mcc4 (TEY of Sample) (by Internal:
mcc1)
#
395.00,395.00,43415.00,77413.00,16810.00,17165.00,3870
0.75,40182.455404,71649.071063,15558.380176,15886.9479
91,35819.213659,1.080447,1.445104,1.000000,1.337506
```

Research Metadata Input dialog box. Fields: Sample Name: Au13_0013_n1s, Sample Formula: , Authors: , Affiliation: , Element: , Edge: , Preparation Method: , Calibrated To: , Temperature: , Scan Mode: , Chamber Conditions: , Comments: . Buttons: OK, Cancel.

```
In [8]: # Saving dataset for analysis with pyMCA
        save_pymca_4d_stack_h5(sgm_data, normalize=True)
```

Detected Grid: 17 x 17

Energies: 182

Normalizing by mcc1...

Building stacks (XRF 3D, XANES 3D, Full 4D Hypercube)...

```
-> Progress: 20/182 energies.
-> Progress: 40/182 energies.
-> Progress: 60/182 energies.
-> Progress: 80/182 energies.
-> Progress: 100/182 energies.
-> Progress: 120/182 energies.
-> Progress: 140/182 energies.
-> Progress: 160/182 energies.
-> Progress: 180/182 energies.
-> Progress: 182/182 energies.
```

[SUCCESS] Multi-mode HDF5 saved to: D:/Data/Testing_jupyter-python-axis2000/N_stack

1. For XRF ROI Imaging: Open 'xrf_measurement/sdd_xrf_stack_3d'
2. For XANES PCA: Open 'xanes_measurement/sdd_xanes_stack_3d' -> 'Load as 1D Stack'
3. For Full Hypercube: Open 'full_4d_measurement'

```
Out[8]: 'D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack/202
        6-04-24_223037_stack_data_PyMca_4D.h5'
```

```
In [3]: # Saving for PCA/CA and for pyMCA analysis
        # Need to select the appropriate roi for the edge of interest
        print("PyMca version", PyMca5.__version__)
        saved_pymca_stack=save_pymca_stack_h5(sgm_data, channel_roi=(34,58))
```

PyMca version 5.9.5

-> No map_roi provided; using full scan range.
-> Target Export ROI: X=[3.752, 5.268], Y=[-2.900, -1.845]
-> Using Internal I0 (mcc1) for normalization.

Processing 182 energy steps for 4 detectors...

-> Progress: 10/182 energies processed.
-> Progress: 20/182 energies processed.
-> Progress: 30/182 energies processed.
-> Progress: 40/182 energies processed.
-> Progress: 50/182 energies processed.
-> Progress: 60/182 energies processed.
-> Progress: 70/182 energies processed.
-> Progress: 80/182 energies processed.
-> Progress: 90/182 energies processed.
-> Progress: 100/182 energies processed.
-> Progress: 110/182 energies processed.
-> Progress: 120/182 energies processed.
-> Progress: 130/182 energies processed.
-> Progress: 140/182 energies processed.
-> Progress: 150/182 energies processed.
-> Progress: 160/182 energies processed.
-> Progress: 170/182 energies processed.
-> Progress: 180/182 energies processed.
-> Progress: 182/182 energies processed.

Successfully saved PyMca-compatible HDF5 stack (multi-detector) to: D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5

Datasets: sdd1, sdd2, sdd3, sdd4, sum, average

Default Signal: average

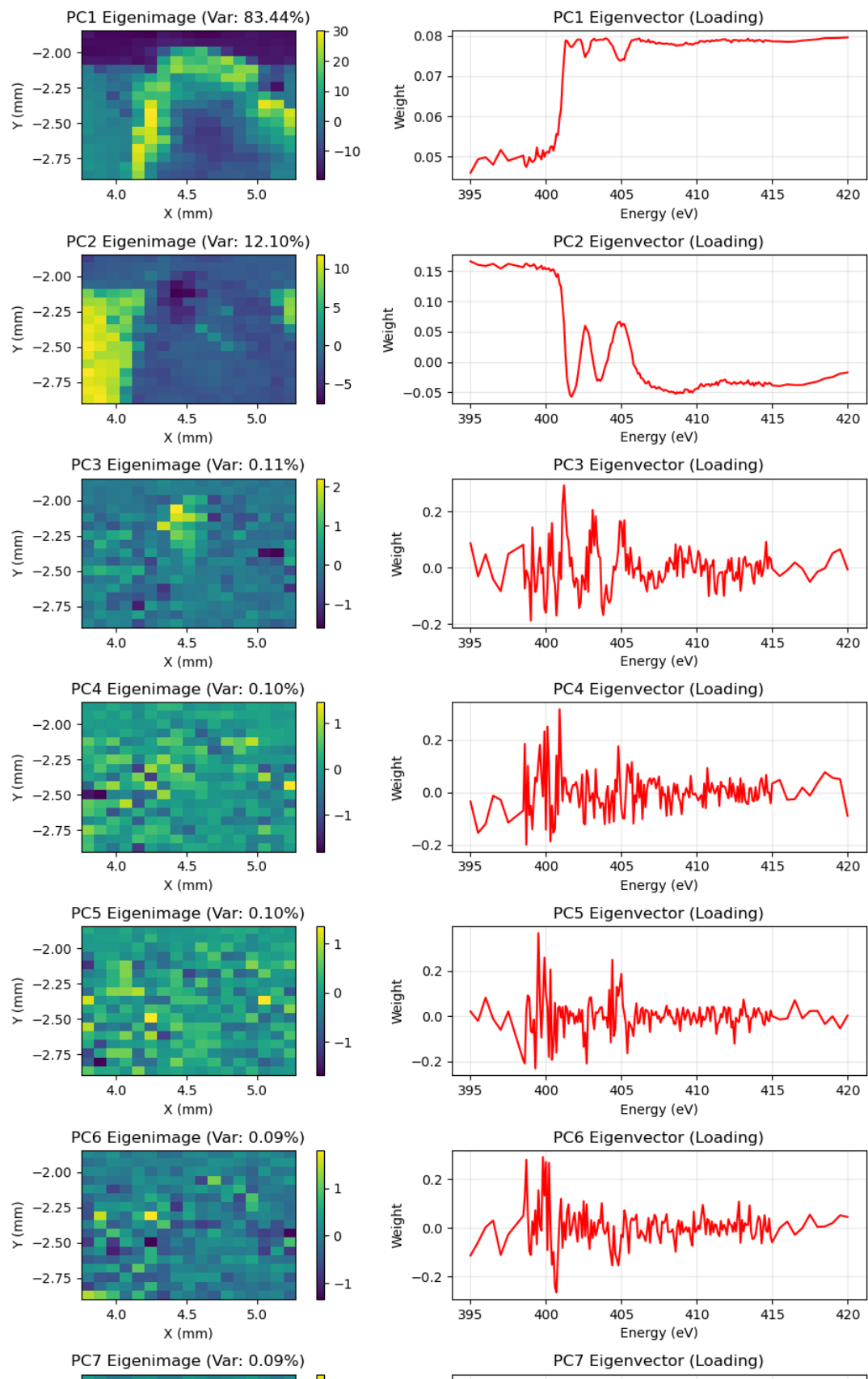
```
In [4]: # For single detector  
# Determined the principal components for each SDD, or for the sum or average of ea  
# Select the SDD and number of components.  
# Displays the eigenimages and vectors  
  
pca_results=pca_xanes_analysis(saved_pymca_stack, dataset_name ='sdd3', n_component
```

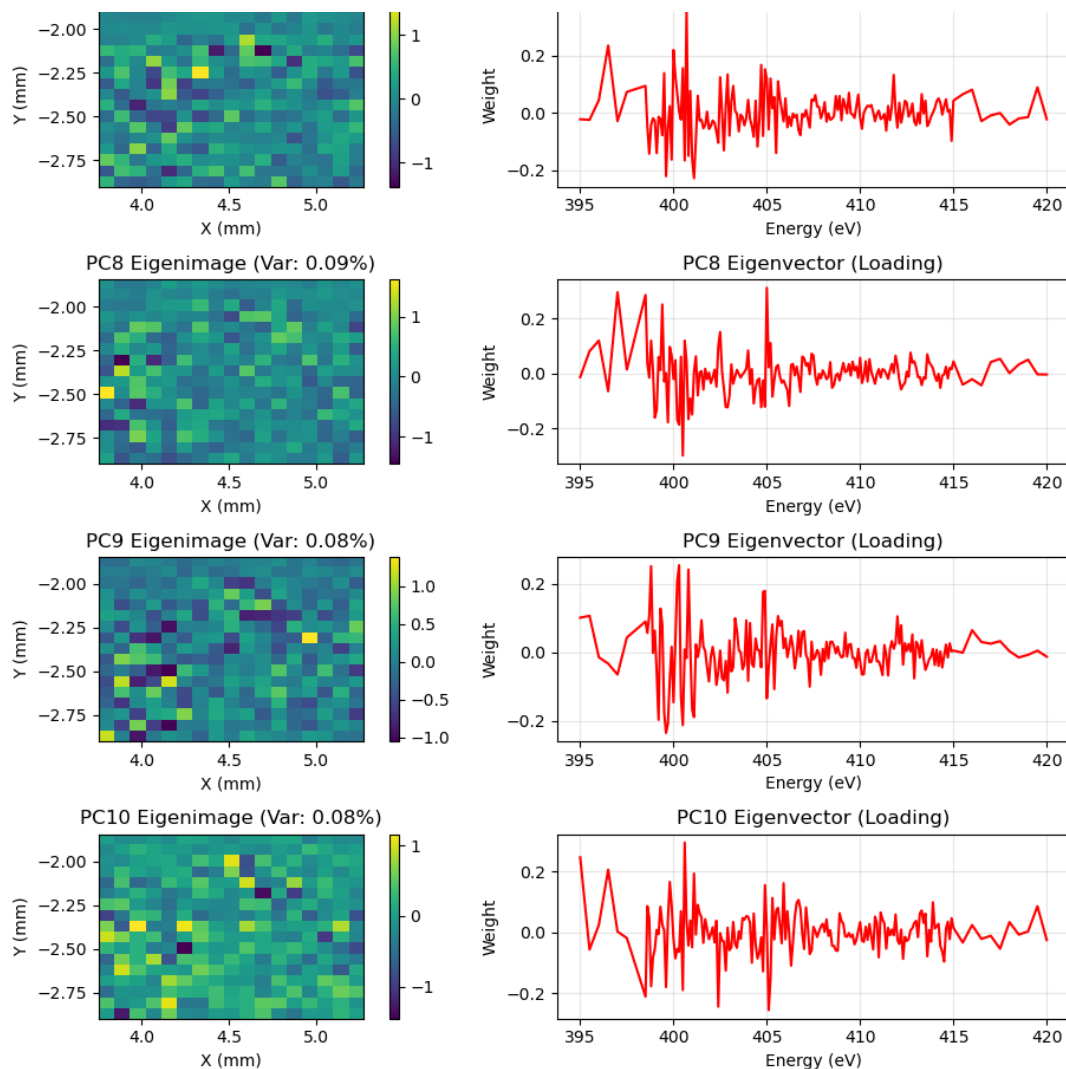
Loading dataset 'sdd3' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...

-> sdd3 stack shape: (17, 17, 182)
-> sdd3 results saved to HDF5.
-> Preview plot saved to: D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_stack\2026-04-24_223037_stack_data_PyMca_sdd3_pca_preview.png

Figure

PCA Results for Stack: sdd3





```
In [5]: # For single detector
# Plots cluster maps and spectra for each detector and number of clusters selected
cluster_results=cluster_xanes_analysis(pca_results, dataset_name='sdd3', use_full_m
```

Loading PCA results for 'sdd3' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...

c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\threadpo
olctrl.py:1226: RuntimeWarning:

Found Intel OpenMP ('libiomp') and LLVM OpenMP ('libomp') loaded at
the same time. Both libraries are known to be incompatible and this
can cause random crashes or deadlocks on Linux when loaded in the
same Python program.

Using threadpoolctl may cause crashes or deadlocks. For more
information and possible workarounds, please see

https://github.com/joblib/threadpoolctl/blob/master/multiple_openmp.md

warnings.warn(msg, RuntimeWarning)

c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.

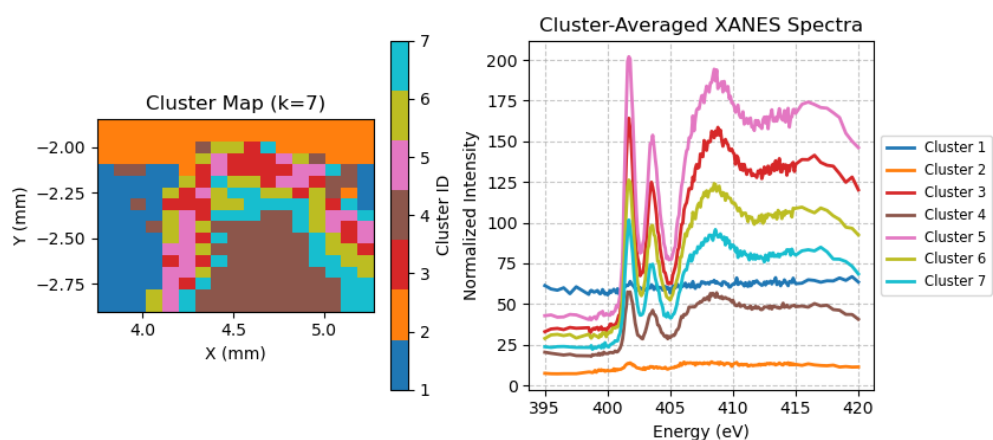
warnings.warn(

-> sdd3 cluster results saved.

-> Preview plot saved to: D:/Data/Testing_jupyter-python-axis2000/N_stacks/22302
9_Au13_0013_n1s_stack\2026-04-24_223037_stack_data_PyMca_sdd3_cluster_preview.png

Figure

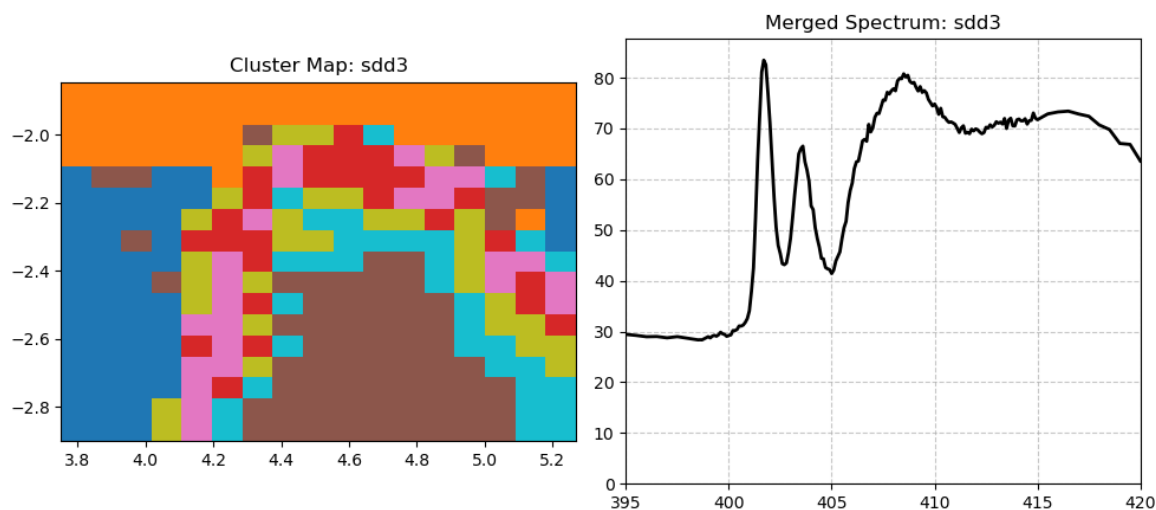
K-Means Cluster Analysis: sdd3



```
In [8]: # Allows for pixels to be exclude from the spectra
interactive_cluster_merger(cluster_results, dataset_name = 'sdd3')
```

VBox()

merger_fig_1989967132304



```
In [8]: # For all detectors
pca_results=run_pca_all_detectors(saved_pymca_stack, n_components=10)
```

```
=====
Running Multi-Detector PCA Analysis
=====
Loading dataset 'sdd1' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_
Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
-> sdd1 stack shape: (17, 17, 182)
-> sdd1 results saved to HDF5.
Loading dataset 'sdd2' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_
Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...

-> sdd2 stack shape: (17, 17, 182)
-> sdd2 results saved to HDF5.

Loading dataset 'sdd3' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_
Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
-> sdd3 stack shape: (17, 17, 182)
-> sdd3 results saved to HDF5.
Loading dataset 'sdd4' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_
Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...

-> sdd4 stack shape: (17, 17, 182)
-> sdd4 results saved to HDF5.

Loading dataset 'average' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/2230
29_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...

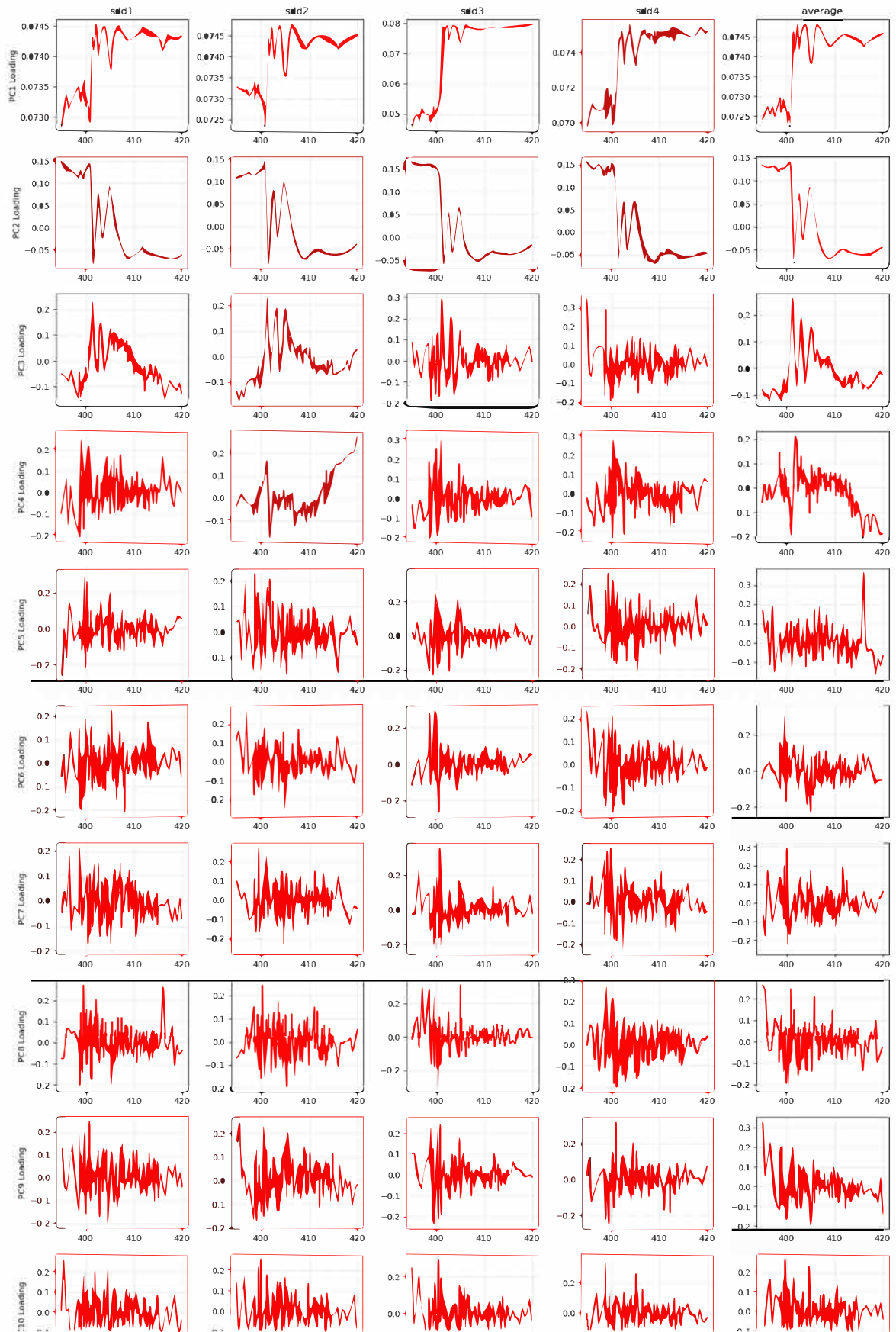
-> average stack shape: (17, 17, 182)
-> average results saved to HDF5.

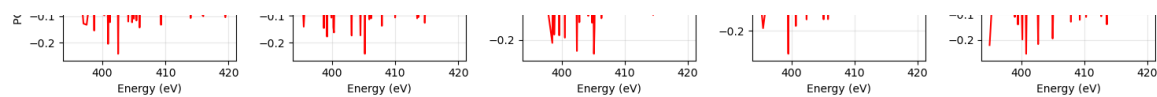
Multi-detector plots saved to:

-> D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_stack\202
6-04-24_223037_stack_data_PyMca_pca_comparison_images.png

-> D:\Data\Testing_jupyter-python-axis2000\N_stacks\223029_Au13_0013_n1s_stack\202
6-04-24_223037_stack_data_PyMca_pca_comparison_vectors.png
```

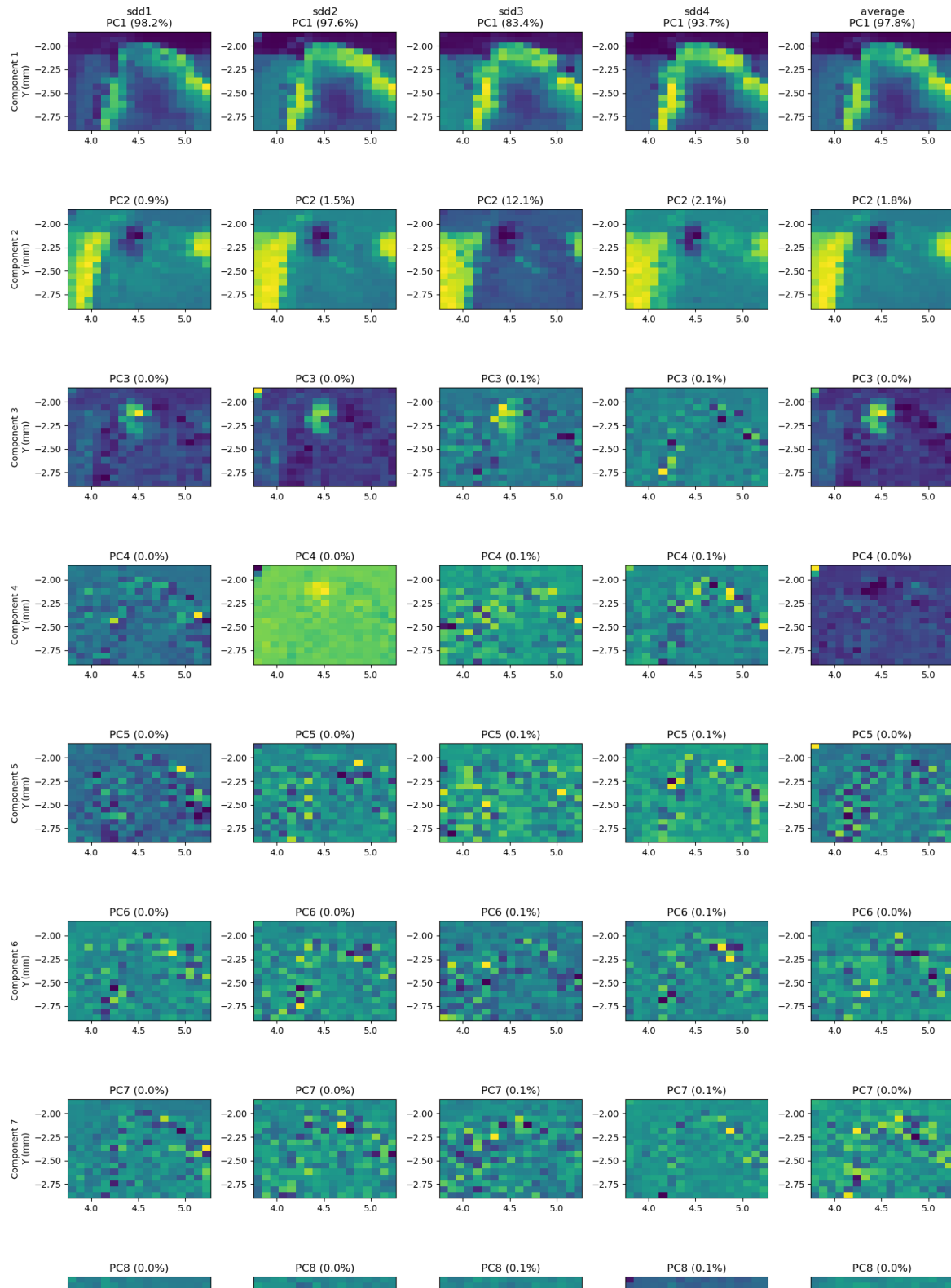
PCA Eigenvector (Loading) Comparison: 2026-04-24_223037_stack_data_PyMca

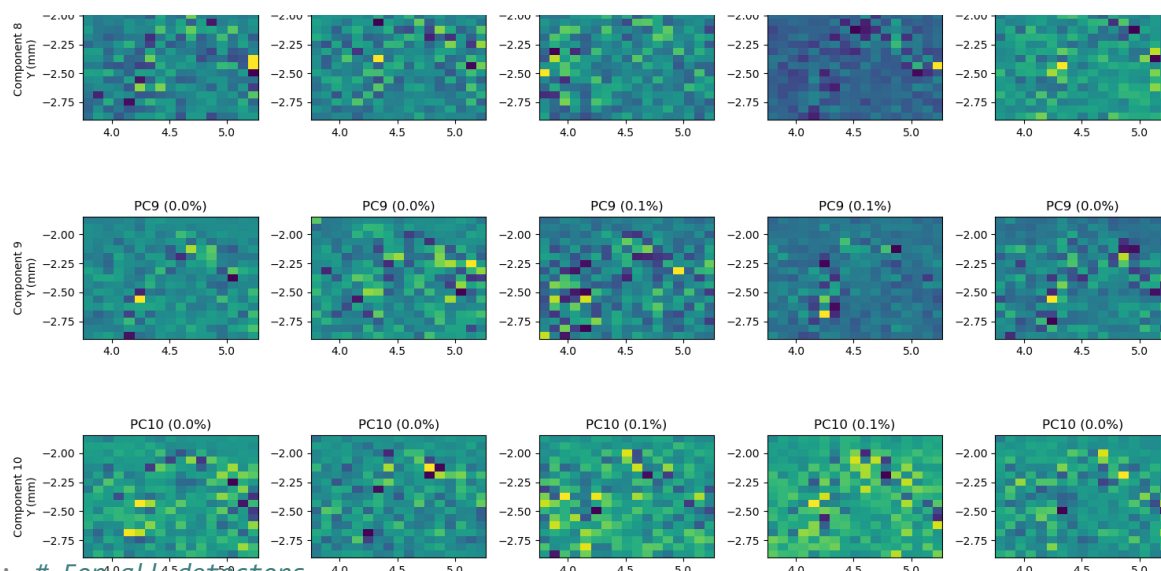




Figure

PCA Eigenimage Comparison: 2026-04-24_223037_stack_data_PyMca





In [9]: *# For all detectors*

```
cluster_results=run_clustering_all_detectors(pca_results, use_full_metadata=False,
```

```
=====
Running Multi-Detector Clustering Analysis
=====
```

```
Loading PCA results for 'sdd1' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
```

```
-> sdd1 cluster results saved.
```

```
Loading PCA results for 'sdd2' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
```

```
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
```

```
warnings.warn(
```

```
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
```

```
warnings.warn(
```

```
-> sdd2 cluster results saved.
```

```
Loading PCA results for 'sdd3' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
```

```
-> sdd3 cluster results saved.
```

```
Loading PCA results for 'sdd4' from D:/Data/Testing_jupyter-python-axis2000/N_stack
s/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...
```

```
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
```

```
warnings.warn(
```

```
c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn
\cluster\_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Wind
ows with MKL, when there are less chunks than available threads. You can avoid it by
setting the environment variable OMP_NUM_THREADS=2.
```

```
warnings.warn(
```

-> sdd4 cluster results saved.

Loading PCA results for 'average' from D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack/2026-04-24_223037_stack_data_PyMca.h5...

-> average cluster results saved.

c:\Users\dynesj\AppData\Local\miniforge3\envs\gemini_cli2\Lib\site-packages\sklearn\cluster_kmeans.py:1425: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=2.

warnings.warn(

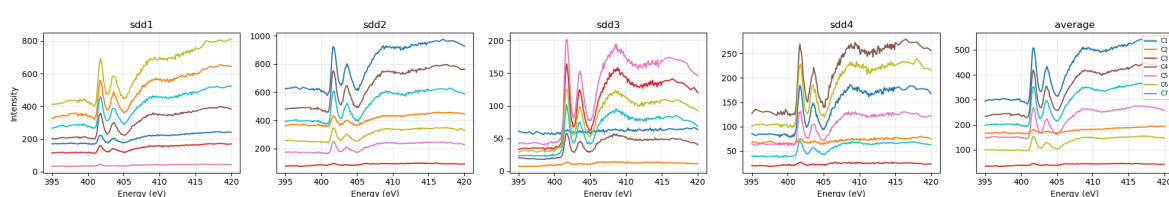
Multi-detector clustering plots saved to:

-> D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack\2026-04-24_223037_stack_data_PyMca_cluster_comparison_maps.png

-> D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack\2026-04-24_223037_stack_data_PyMca_cluster_comparison_spectra.png

Figure

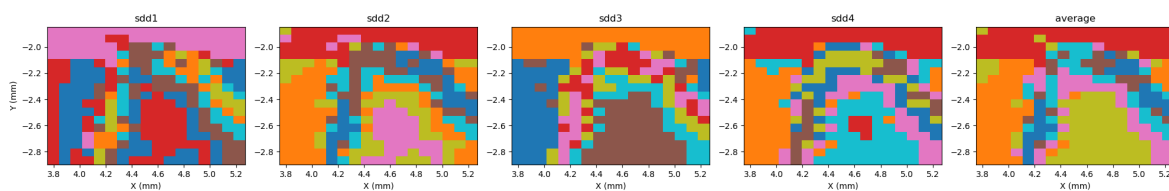
Multi-Detector Cluster Spectra: 2026-04-24_223037_stack_data_PyMca



-> Combined cluster sums saved to: D:/Data/Testing_jupyter-python-axis2000/N_stacks/223029_Au13_0013_n1s_stack\2026-04-24_223037_stack_data_PyMca_all_detectors_cluster_sums.csv

Figure

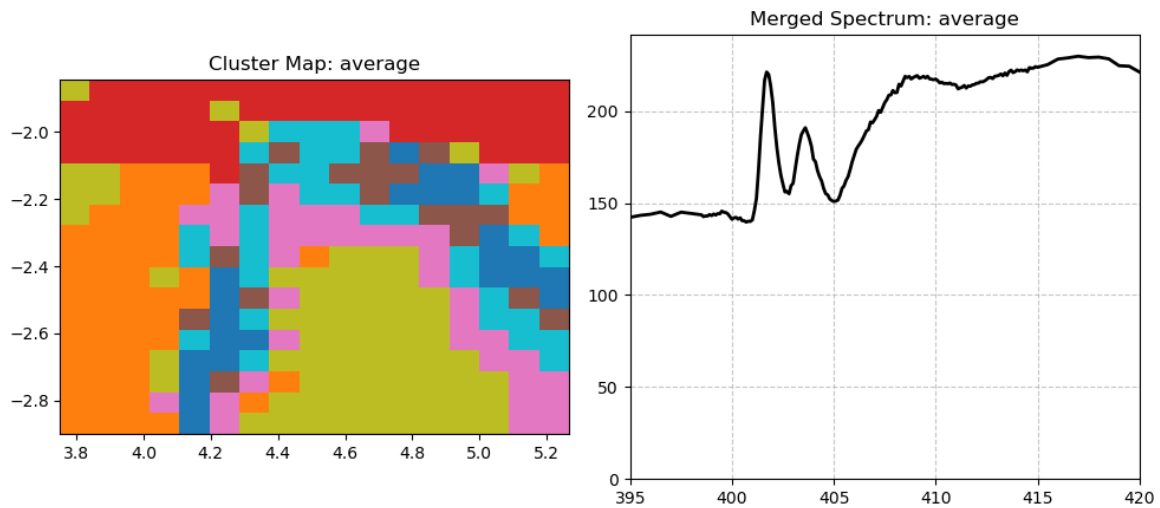
Multi-Detector Cluster Maps: 2026-04-24_223037_stack_data_PyMca



```
In [11]: #For all detectors
         interactive_cluster_merger(cluster_results)
```

VBox()

merger_fig_1989967132304



In [3]: `# Adds the I0 from the Au mesh (mcc1) from a number of N1s spectra
The reason is to increase the S/N ratio of the I0, thus increasing the S/N of the
sum_mcc1_spectra()`

Loading 2026-04-24_223037_stack_data.h5...

Loading 2026-04-24_205859_stack_data.h5...

Loading 2026-04-24_195642_stack_data.h5...

[SAVE SUCCESS] MCC1 Sum/Average saved to:

--> D:\Data\Testing_jupyter-python-axis2000\N_stacks\N-mcc1-from-N-stacks1.csv

In [

